Language Models are Unsupervised Multitask Learners

Radford, Wu, Child, Luan, Amodei and Sutskever (OpenAI, 2019), the GPT-2 paper

Paper summary for the CS-781 capstone

The paper at a glance

Citation

A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, I. Sutskever. *Language Models are Unsupervised Multitask Learners*. OpenAI technical report, 2019.

One-sentence summary

A large decoder-only language model (GPT-2, 1.5B parameters) trained on diverse web text starts to perform tasks such as reading comprehension, summarisation and translation **zero-shot**, with no task-specific training.

Why it matters for our capstone

Our A1 baseline is a GPT-2-style decoder (pre-LN, 1,024 context, scaled residual init), and the starter notebook uses the GPT-2 tokenizer. Phase 1, continuing a legal prompt, is exactly GPT-2's core task.

Where it appears in our slides

Main deck: "A1: decoder-only Transformer". Glossary: "Byte-level BPE" and "GPT-2 tokenizer".

Roadmap

1. The problem
2. The basic idea
3. Results
4. Assessment and outlook

Narrow experts vs general systems

- The usual recipe: collect a labelled dataset for *one* task, train, test on similar data.
- This gives **narrow experts**: brittle when inputs or task framing shift slightly.
- Multitask learning helps, but the largest efforts used only 10–17 (dataset, objective) pairs. From a meta-learning view, that is only 10–17 *training examples* of "tasks".
- Creating labelled datasets at the scale needed would be very hard.

Question: can a language model learn tasks *from raw text alone*?

How transfer had evolved

1. Word vectors fed to task-specific models (word2vec, GloVe).
2. Contextual representations from RNNs (ELMo).
3. Transfer whole self-attention stacks, then fine-tune (GPT, BERT).
4. **This paper**: no fine-tuning at all. The pre-trained model does the task directly, zero-shot.

Tasks are just text, so a language model can learn them

A language model estimates

$$p(x) = \prod_{i=1}^{n} p(s_i \mid s_1, \ldots, s_{i-1})$$

A general system should model $p(\text{output} \mid \text{input}, \text{task})$. In language, the task itself can be written as text:

- (*translate to french*, English text, French text)
- (*answer the question*, document, question, answer)

The supervised objective is the *same* as the language-model objective, only evaluated on part of the sequence. So a good enough LM should pick up tasks that appear naturally in text.

Natural demonstrations in WebText

The paper found translation examples appearing naturally on web pages, e.g.:

"I'm not the cleverest man in the world, but like they say in French: Je ne suis pas un imbecile [I'm not a fool]."

To predict such text, the model must in effect learn to translate.

Legal analogy

Judgments contain "In short, . . . " summaries and quoted statutes, so a legal LM may learn these skills from judgments alone.

Zero-shot prompting: invoking a task with text

The task is signalled only by how the input is written. No weights change.

Task	Prompt format used in the paper	Model output
Summarisation	*article* `TL;DR:`	summary text
Translation	`english sentence = french sentence` (a few pairs), then `english sentence =`	French sentence
Reading comprehension	document + conversation history + `A:`	answer
Question answering	example Q/A pairs, then a new question	short answer
Cloze (CBT)	score each of 10 candidate words; pick the most probable sentence	best candidate

Legal example of the same idea

"*. . . the appeal is accordingly dismissed with costs.* `TL;DR:`" → the model continues with a short summary of the judgment, with no summarisation training. This is how Phase 6 could be attempted before any fine-tuning.

WebText: a large, diverse, curated dataset

- Prior LMs trained on one domain: news, Wikipedia or books.
- Common Crawl is huge but full of unintelligible documents.
- **Idea**: use human curation as a quality filter. Scrape every outbound link from **Reddit** with at least 3 karma.
- Result: about **8 million documents, 40 GB** of text, from 45 million links (up to December 2017).
- All Wikipedia pages removed, to avoid overlap with test sets.

Why diversity matters

The more varied the text, the more naturally occurring "task demonstrations" the model sees: Q&A threads, translations, summaries, lists.

Our Phase-1 analogue

The capstone corpus mixes four legal sources (Acts, High Courts, Supreme Court, Tribunals). Mixing ratios are our lever for diversity.

Input representation: byte-level BPE

Goal: assign a probability to *any* string, with no lower-casing, no `<unk>`.

- Pure byte-level LMs were not competitive; Unicode-level BPE needs a base vocabulary of over 130,000.
- **Byte-level BPE**: base vocabulary of only **256 bytes**; merges learned on top. Final vocabulary: **50,257**.
- **Fix**: plain BPE learned many wasteful variants such as `dog.`, `dog!`, `dog?`. So merges across character categories (letters vs punctuation) are forbidden, except for spaces.

Easy example

Without the rule, the vocabulary may hold `dog`, `dog.`, `dog!`, `dog?`, `dog,`: five slots for one word.

With the rule: `dog` plus a separate `.`, `!`, `?`. The saved slots go to other words.

Legal version: `Section`, `Section,`, `Section.` collapse into one token `Section` plus punctuation.

In our code

Our tokenizer uses the same byte-level BPE design, trained on legal text.

Model: a scaled-up GPT

Parameters	Layers	d_{model}
117M	12	768
345M	24	1024
762M	36	1280
1542M (GPT-2)	**48**	**1600**

The smallest equals the original GPT; the second equals BERT-large in size.

Changes from GPT:

- **Pre-LN**: LayerNorm moved to the *input* of each sub-block, plus one final LayerNorm after the last block.
- **Scaled residual init**: residual-layer weights scaled by $1/\sqrt{N}$, where N is the number of residual layers.
- Vocabulary 50,257; context 512 $\rightarrow$ **1,024** tokens; batch size 512.

Same choices in our A1

Pre-LN, final LayerNorm, 1,024 context, tied embeddings, residual init scaled by $1/\sqrt{2L}$.

Zero-shot language modelling

	LAMBADA (PPL)	LAMBADA (ACC)	CBT-CN (ACC)	CBT-NE (ACC)	WikiText-2 (PPL)	PTB (PPL)	enwik8 (BPB)	WikiText-103 (PPL)
Previous SOTA	99.8	59.23	85.7	82.3	39.14	46.54	0.99	18.3
117M	35.13	45.99	87.65	83.4	29.41	65.85	1.16	37.50
345M	15.60	55.48	92.35	87.1	22.76	47.33	1.01	26.37
762M	10.87	60.12	93.45	88.0	19.93	40.31	0.97	22.05
1542M	**8.63**	**63.24**	93.30	**89.05**	**18.34**	**35.76**	**0.93**	**17.48**

Reading it

No training on any of these datasets. GPT-2 sets a new state of the art on **7 of 8** language-modelling benchmarks. It fails only on One Billion Word (42.16 vs 21.8 perplexity), whose sentence-level shuffling destroys the long-range structure GPT-2 relies on. Every metric improves as the model grows.

Long-range dependencies: LAMBADA and CBT

LAMBADA: predict the last word of a passage, which humans can only guess with at least 50 tokens of context.

- Perplexity: 99.8 $\rightarrow$ **8.6**.
- Accuracy: 19% (previous LMs) $\rightarrow$ 52.66%, and **63.24%** with a stop-word filter.
- Most errors were valid continuations, but not valid *final* words.

Children's Book Test: choose the right word among 10 candidates. New records: **93.3%** common nouns, **89.1%** named entities.

Legal version of LAMBADA

"... The High Court had convicted the appellant under Section 302. Having examined the evidence, we find it unreliable. The conviction under Section 302 is therefore ____"

The right word, "set aside", needs the whole passage, not just the last few words. This is the skill our 512-word prompts test.

Zero-shot tasks

Task	GPT-2 result	Comparison
CoQA reading comprehension	**55 F1**	matches or beats 3 of 4 supervised baselines (humans: 89)
Winograd Schema	**70.70%**	+7 points over the previous best
WMT-14 Fr→En	11.5 BLEU	best unsupervised MT: 33.5
WMT-14 En→Fr	5 BLEU	slightly worse than word-by-word substitution
Natural Questions	4.1% exact match	5.3× the smallest model; retrieval systems reach 30–50%

Surprises

CoQA used none of its 127,000+ training question–answer pairs. For translation, non-English pages had been filtered out: only about 10 MB of French remained, 500× less than usual. On Natural Questions, accuracy is 63.1% on the 1% of questions GPT-2 is most confident about, so its confidence is well calibrated.

Summarisation with “TL;DR:”

CNN/Daily Mail, ROUGE F1:

Method	R-1	R-2	R-L	Avg
Bottom-Up Sum (SOTA)	41.22	18.68	38.34	32.75
Lede-3	40.38	17.66	36.62	31.55
Seq2Seq + Attention	31.33	11.81	28.83	23.99
GPT-2 TL;DR:	29.34	8.27	26.58	21.40
Random 3 sentences	28.78	8.63	25.52	20.98
GPT-2, no hint	21.58	4.03	19.47	15.03

Top-k sampling with $k = 2$; the first 3 generated sentences are the summary.

Reading it

- Only just above picking 3 random sentences.
- Removing “TL;DR:” drops the average by **6.4 points**: the text hint really does switch on summarisation behaviour.
- Errors: focuses on recent content, confuses details such as how many cars crashed.

For Phase 6: zero-shot is a baseline to beat, not a solution. Fine-tuning is needed.

Generalisation vs memorisation

Could GPT-2 just be recalling test data it saw in WebText?

- Bloom filters of all WebText 8-grams measure test-set overlap.
- Benchmark test sets overlap WebText by 1–6% (average 3.2%). They overlap *their own training splits* more: 5.9% on average (1BW: 13.2%).
- Removing overlapping LAMBADA examples: perplexity 8.6 $\rightarrow$ 8.7, accuracy 63.2% $\rightarrow$ 62.9%. A small effect.
- Train and held-out WebText perplexity fall **together** with size: even GPT-2 still **underfits**.

A lesson we apply

The authors recommend n-gram overlap checks when building train/test splits.

Our Phase-1 pipeline splits train and validation **by document**, so no dev prompt comes from a training judgment. An 8-gram overlap check between our dev prompts and training bins would be a useful extra report item.

What GPT-2 means for our capstone

GPT-2 finding	Implication for us
Zero-shot ability grows log-linearly with size	Within 50–300M, bigger A1 variants should continue text better
Still underfits at 1.5B	Our 80M models on about 200M tokens are surely underfit; more tokens should help
Diverse data creates task demonstrations	Mixing all four legal sources may help more than one source
Task hints ("TL;DR:") switch on behaviour	Try zero-shot prompts as baselines for Phases 3–6
Top-k sampling for generation	The origin of the decoding knobs we tune for Phase 1
Data-overlap checks	Split by document; report n-gram overlap

Strengths and limitations

Strengths

- Showed zero-shot task transfer from pure language modelling.
- Clear scaling evidence over four model sizes.
- Byte-level BPE: any string can be modelled and evaluated.
- Careful memorisation analysis.
- Simple, influential architecture recipe.

Limitations (several stated by the authors)

- Zero-shot performance is "still far from usable" in practice.
- Summarisation, translation and QA are rudimentary.
- Unidirectional: may be less efficient than BERT for understanding tasks.
- Evaluation of many tasks is noisy; some tiny (Winograd: 273 examples).
- Technical report; limited training details.
- Web data brings biases and possible misuse, which led OpenAI to a staged release.

Future research directions

Stated by the authors

- Find the ceiling with **fine-tuning**, e.g. on decaNLP and GLUE.
- Test whether more data and capacity overcome the inefficiency of **unidirectional** representations shown by BERT.
- Evaluate more tasks; better de-duplication.

What followed

- GPT-3 (Brown et al., 2020): 175B parameters and **few-shot in-context learning**.
- Scaling laws; instruction tuning and RLHF (Kaplan et al., 2020; Ouyang et al., 2022).

Open questions

- *Why* do tasks emerge from next-token prediction?
- How much of zero-shot skill is memorisation?
- How small can a model be and still show useful zero-shot behaviour?
- For legal NLP: does an 80–300M legal model show *any* zero-shot skill, e.g. answering "The statute applicable here is" correctly? Measuring this would be a novel Phase-1 analysis.

Summary

- Tasks can be written as text, so a language model trained on diverse text can learn them without supervision.
- **WebText**: 40 GB from Reddit-curated links. **Byte-level BPE** with a 50,257 vocabulary and a no-cross-category merge rule.
- **GPT-2**: 1.5B-parameter decoder with pre-LN, scaled residual init and a 1,024-token context.
- Zero-shot: state of the art on 7 of 8 LM benchmarks; CoQA 55 F1; LAMBADA 8.6 perplexity; weak but real summarisation and translation.
- Performance grows log-linearly with size; even GPT-2 underfits.
- For us: the template for A1 and for Phase 1's task of continuing a prompt.

Reference

Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., and Sutskever, I. (2019). Language Models are Unsupervised Multitask Learners. OpenAI technical report.

Other work cited: Radford et al. (2018) GPT; Devlin et al. (2018) BERT; McCann et al. (2018) decaNLP; Sennrich et al. (2016) BPE; Fan et al. (2018) top-k sampling; Paperno et al. (2016) LAMBADA; Reddy et al. (2018) CoQA; Brown et al. (2020) GPT-3; Kaplan et al. (2020) Scaling Laws; Hoffmann et al. (2022) Chinchilla; Ouyang et al. (2022) InstructGPT.